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The Perception of Fidelis Senior High Technical–Vocational and Livelihood Information and Communication Technology Students on the Importance of Coding Skills for Web Development

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**THE PERCEPTION OF FIDELIS SENIOR HIGH TECHNICAL-VOCATIONAL
AND LIVELIHOOD INFORMATION AND COMMUNICATION
TECHNOLOGY ON THE IMPORTANCE OF CODING SKILLS
FOR WEB DEVELOPMENT**

**A Research Presented to the Fidelis Senior High School Department
First Asia Institute of Technology and Humanities
2 Pres. Laurel Highway, Tanauan City Batangas**

**In Partial Fulfillment of the Requirements for Senior High School Program,
Technical, Vocational and Livelihood – ICT Strand**

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December 2020

DEDICATION

The researchers dedicate this study to their family for the support that they are giving to them, not only for the financial support but also in terms of moral support, especially to Mr. Oswald Zach Abutal, who selflessly gave knowledge to help the researchers to construct their paper. Also, the researchers dedicate this research study to their friends who help and come together with them in doing this research. To Ms. Macy Perez for the continuous support that she gave to them, for giving them motivations to do their best for this research and for giving them inspirations to learn the values of conducting research not just by know-how. And lastly, the researchers highly dedicate this research study to Almighty God who always support us from the beginning up to the end of this research study.

ABSTRACT

The descriptive quantitative study aims to determine the perception of Fidelis Senior High School Technical Vocational and Livelihood Information and Communication Technology (TVL ICT) students on the importance of coding skills for web development. In terms of choosing the skills or competency for web development, the study also conducted to identify the least considered coding skills of TVL ICT students. Furthermore, it also identify the frequency of how often the respondents practice their coding skills equivalent to how they perceive the importance of coding skills for web development. In addition, the respondents of the study were consist of fifty (50) students both from the total population of Grade 11 and Grade 12 TVL ICT students. The chosen procedure for the entire population of the respondents is particularly considered as a total population sampling method. In addition, the study made use of descriptive survey research methodology. By using an adapted questionnaire with modification, the data from the respondents were gathered. Particularly, the adapted questionnaire contains a 5-point Likert Scale Table is used as a rating scale to measure the perception of the respondents. Lastly, this is followed by interpreting the gathered data using the statistical treatment of weighted mean to analyse the data.

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

This chapter presents the introduction, statement of the problem, significance of the study, scope and delimitations, theoretical and conceptual framework, and definition of terms.

Introduction

Technology is one of mankind's greatest invention. It helps individuals to live life much more easier, especially in this digital age ("Mankind", n.d.). It comes in many different forms but one of which is the Internet. The internet has provided global access to data and communication. As a technology, the Internet also aided a lot of people, particularly businesses and industries since it created plenty of opportunities for them by making the world a little bit smaller and connected (Gleeson, 2019). As a business, one should have his/her own site to be able to use the Internet, similar to a business that also needs its own place in a market. Furthermore, to access the Internet a bit more comprehensible, the owners should create an attractive and user-friendly interface. Not only that, one should also consider the system in order to efficiently operate its business online. With all these matter at hand, it is safe to say that using the Internet is a tough work. However, this is where the web development industry enters the stage. Moreover, the web development industry, in this digital age, has become one of the most successful businesses in the modern world ("Web Development", n.d.). It has made a place where information is accessible for anyone. Web development is a work involved in developing websites for the

Internet. It has helped companies and businesses to further spread their reach. Web development includes aspects such as web design, web programming, web publishing, and database management. All these aspects are the duties of a web developer. Furthermore, among web developers, “Web Development” typically refers to the core aspects of building websites: writing codes (“10 Skills”, 2014). Coding or programming is defined as a structure of symbols and rules used to denote commands. At its core, coding is a problem-solving activity and a form of expression. In order to act efficiently in the web development industry, you need to learn specific sets of skills called computer programmer skills or “coding skills”. In fact, as with various other fields, coding skills can also be clustered into “hard skills” and “soft skills” (“10 Skills”, 2014). Hard skills or technical skills are denoted as all the *hands-on* work, learning, knowledge and abilities from practicalities. In contrast, soft skills are the social or interactive skills needed in collaboration with other people. Acquiring these skills are important since it will help you perform tasks efficiently and to function well in the web development industry. In addition to that, these coding skills should be earnestly practiced and maintained for it will train you to be a competent web developer. In web development industry, competency in coding skills becomes very significant to those individuals who pursue jobs and promotions (“Web Development”, n.d.). For it is one of the major requirements when entering certain jobs. As this industry grows and increases in complexity, it demands more competent individuals in their workplace. This shows that preparation and training for aspiring web developers are in need of provision with this matter. Furthermore, in order to respond competently to industry prerequisites, it is noteworthy that web development education should provide accommodations in order to satisfy these needs. At the early stages, universities were

insightful with this concern having them to provide courses related to programming to sustain this matter (Sabarinath, 2019). As the solution resolves the matter, many institutions obtain these solutions to prepare students, on both hard skills and soft skills of coding skills. In this way, aspiring students who will join the web development industry will aid them to acquire their desired careers and develop their specific skills and abilities needed in this industry (Sabarinath, 2019).

Since the career path for web development is part of Fidelis Senior High School Technical Vocational and Livelihood Information and Communication Technology track, this study aims to know the perception of Fidelis Senior High School Technical Vocational and Livelihood Information and Communication Technology students on the importance of coding skills for web development which will help the students to improve their coding skills by knowing their perception and how they perceived it. Through that process, the student-researchers can give a good account for enhancement activities or any solution for improvement on their coding skills of Fidelis Senior High School Technical Vocational and Livelihood Information and Communication Technology students for web development.

Statement of the Problem

This study aims to identify the TVL-ICT student's perception on the importance of coding skills for web development. More specifically, the student-researchers seek answers from the following questions:

1. How do the TVL-ICT students perceive the importance of coding skills for web development in terms of:

1.1. Code Analysis;

1.2. Programming Languages;

1.3. Design;

1.4. Communication;

1.5. Self-reliance; and

1.6. Empathy?

2. Which among the coding skills for web development is least considered by TVL-ICT students?

3. If coding skills are important, how often does the respondent practice his/her coding skills as a TVL-ICT student?

Significance of the Study

In this section, this is where the student-researchers, enumerate the study's purpose and significance and explains the study's benefits (Strawberry Fields, 2012). As the study learns the perception of the respondents on the importance of coding skills for web development, the results give a good account for the following beneficiaries.

TVL-ICT Students. This study will help them to improve their coding skills. Also, it will provide them knowledge on how they could improve such skills.

Instructor of Web Development. Knowing the perceptions of the students on the importance coding skills will lead them on supporting and teaching the students in more objective and careful way that results for more understanding of students' applications and improvements.

Parents. The acquired information from the study will help the parents to support their child's necessities on perceiving such skills.

School Community. The outcome of the study will also help the school on giving good accounts for enhancement activities for aspiring web developers.

Future Researchers. For those upcoming researchers, this study can serve as their guide or reference on conducting a research study that relates to their chosen topic.

Scope and Delimitations

The scope of the study primarily signify every contents that will be included in the study ("Scope of the Study," 2014). The study concentrates only on the perception of the Fidelis Senior High School TVL ICT students on the importance of coding skills for web development. The student-researchers used a descriptive-survey research design and arranged survey questionnaires to get the data for the study. The research will be conducted at Fidelis Senior High School of First Asia Institute of Technology and Humanities in the

first term of the year 2020-2021. The chosen respondents for this study are the total population of the Technical Vocational and Livelihood Information and Communication Technology strand, both Grade 11 and 12 of Fidelis Senior High School. With the numbers of fifty (50) students exception to us student-researchers. On the contrary, the delimitation of the study are the set boundaries in which prepared by the researchers for them to regulate the range of a study (Franklin, 2016). The study does not cover areas of web development such as essential skills and professional skills. Also, the study does not cover other areas of Web Development such as database management, frameworks, algorithms and its paired programming languages. Lastly, the study does not include other non-academic track.

Theoretical Framework

The backbone of this study is guided by Social Intelligence Theory (Thorndike, 1920), and the Multiple Intelligence Theory (Gardner, 1983) and Cognitive Flexibility Theory (Spiro, 1990).

Thorndike's (1920) Social Intelligence Theory (SI) states that it is the ability to capably communicate with others, and to get them to cooperate with you. Sometimes denoted as "people skills", SI includes an attentiveness of situations and the social dynamics that administer individuals, and an understanding of interaction strategies and styles that can help an individual attain his or her objectives in communicating with others. It also includes a certain extent of self-consciousness and insight of one's own perceptions and reaction patterns. Adding to that, in relation to the study's research problem, the Social Intelligence Theory helps to support the soft skills. Since this theory tackles one's own

perception and how it tends to capably communicate with others, it would certainly help this paper on its pursue to identify the respondents perception on the importance of coding skills for web development.

The Multiple Intelligence Theory proposed by Gardner 1983 (as cited in David, 2014) indicates that there are multiple types of human intelligence, each representing different ways of processing information. It includes the bodily-kinesthetic intelligence which entails a person's capability to process information physically through hand and body movement, control, and expression. It also states that visual-spatial intelligence allows individuals to comprehend figures and other types of graphical information. Furthermore, the Multiple Intelligence Theory supports the hard skills of coding skills needed for web development. Also, it explains that individuals have different capabilities in perceiving and processing information which, in relating to the study's research problem, would impact an individual's perception on how he or she perceives the hard skills of coding skills for web development.

The Cognitive Flexibility Theory proposed by Spiro (1990) concentrates on the nature of learning in complex and sophisticated domains. It also indicates that by cognitive flexibility, it is defined as the capability to spontaneously restructure one's knowledge, in various ways, in adaptive response to drastically changing situational demands. Furthermore, this theory helps to support this paper in relation to the one of the parameters which is the proficiency of learning different programming languages that is necessary and has a significant value for one of the hard skills of coding skills for web development. Furthermore, it indicates one's own capability to adapt to various changes, or in relation to this study, is the skill to learn different programming languages for web development.

Conceptual Framework

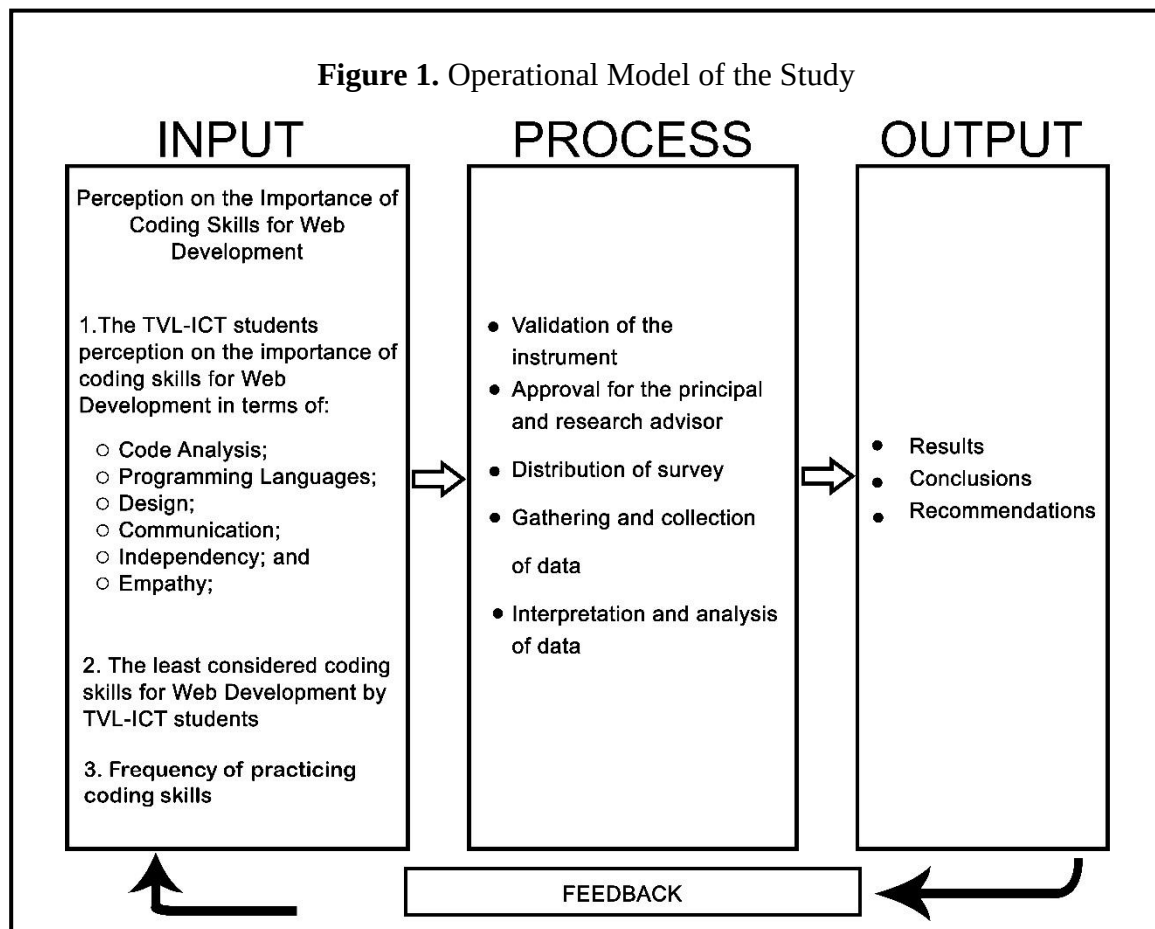


Figure 1. Conceptual Paradigm

The Conceptual Framework provides visuals to the procedure of the operations of the study. It is used to identify the perception of TVL ICT students on the importance of coding skills for website development in Fidelis Senior High School by using the “IPO” model, namely; Input, Process and Output. For the input stage, it shows the specific research questions that we, student-researchers seek to be answered. Next, for the process stage, it shows the process of sending out the questionnaires, the gathering of data and the

interpretation for the research problems to have the fundamental information for the output stage. Lastly, the output stage shows the results and findings of the study where in that process the student-researchers may gather the information and understand the perception of TVL-ICT students.

Definition of Terms

In this part, definition of terms shows how a definition will be used in the study (Kirkpatrick, 2017).

Code Analysis. It is the analysis of source code that is done without actually executing programs. It involves manual detection of vulnerabilities and functional errors in deployed or being deployed software (“Code Analysis”, n.d.).

Coding Skills. It is the process of creating directions and instructions for computers using programming languages (Weinstein, 2020).

Communication. It is the series of transferring information and common understanding from one person to another (Keyton, 2011). In this study, it is one of the soft skills of coding skills for website development.

Design. In the context of website development, it is the planned structure of a website for the implementation of an activity or process (Wikipedia, 2020).

Empathy. It is the competency to share other people’s feelings and experiences and the feelings that a certain person understand (“Empathy,” 2018). With this study, it was defined as one of the soft skills of coding skills needed for web development.

Hard Skills. These are particular, teachable competencies that can be distinguish and calculated, such as typing, inscription, solving, reading and the ability to operate software programs (“Hard Skills,” 2018). In this study, it is defined as all the hands-on work, knowledge, and learning and experiences from practicalities.

Perception. It is the how a person thinks about and understands someone or something (“Perception,” 2018). This is what us, student-researchers seek to find improvements on coding skills for web development.

Programming. Programming is a way on how to instruct the computer to perform various and simultaneous tasks (Goel, 2020).

Programming Language. It is defined as a formal language that includes a set of instructions that produce various kinds of outputs (Wikipedia, 2020).

Self-reliance. It is defined as the ability to do things and make decisions by yourself, without the aid of other people (“Self-reliance”, n.d.).

Soft Skills. It is the individual’s interpersonal competencies and attributes that characterize individual's relationships with others (“Soft Skills,” 2018).

Technology. It is the accumulation of the ways in which social groups provide themselves with the material objects for the advancement of their civilization (“Technology”, 2011).

Web Developer. A web developer is a programmer who specializes in building websites for the Internet (Wikipedia, 2020).

Web Development. It refers to building, creating, and maintaining websites. It includes aspects such as web publishing, web design, web programming, and database management (“Web Development”, n.d.).

Web Development Industry. It consists of individual web developers, web design template companies, custom website design and development services, web domain and hosting sales, and audio and video coding for web development (“Web Design & Development”, n.d.).

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents the topics related studies, literature and its synthesis. The accumulated readings are allocated into two categories: the conceptual literature conferring to the context of the research with its variables and the research literature which states the reliable studies that has a linking to the research topic. Lastly, to connect all these, a synthesis is presented at the end of this chapter.

Conceptual Literature

Since coding skills are now prevalent to have in this modern age, literatures and studies about this competency grow in numbers. This section presents the conceptual literature that discusses the related gathered data from published articles, books, journals, blogs, websites, magazines, novel, and poetry of which have presented the importance of coding skills.

Coding Skills. At its core, coding is an art of expression and solving problems (“10 Skills”, 2014). Besides its existential value, learning and studying how to code competently will provide you countless and advantageous job opportunities. Furthermore, Lee (2010) also stated that these skills; code analysis, programming languages, design, communication, self-reliance and empathy will help you in your endeavour as a coder, even in life as well. In conclusion, mastering coding skills are very important in the web development industry since these skills are the ones that will help you land jobs much more efficiently.

Code Analysis. According to Coding (n.d.), code analysis is the skill of effectively analysing source codes that could be performed without actually executing programs. This involves the process of detecting vulnerabilities and functional errors in deployed or soon-to-be deployed software. This skill requires adequate attention in order to harbour the maximum capability of an individual. Adding to that, this skill helps individual programmers to be competent in their work by proficiently and efficiently debugging codes from source codes or even when analysing their own codes. Lastly, coding analysis is one of the example of coding skills that are essential for if an individual is to pursue a web development job (Lee, 2010).

Programming Languages. One of the computer programming competency required for a sustainable employment is the proficiency in programming languages (Ibezim, 2017). Learning different programming languages is one of the significant determinants of success if one is to enter the web development industry. There is in fact a ton of programming languages that have already been made and most of them are being actively used worldwide. These languages introduce tasks and instructions for programmers to code in order to make a software. Lee (2010) added that proficiency on programming languages of programmers contributes to their overall coding skills for web development that would be essential in undertaking tasks and performing rigorous labors.

Design. According to Chapman (n.d.), programmers who has the knowledge and skill for designing are crucial and play a role in every website building, app developing and product placements. A programmer designer usually focuses in UX Design and UI Design. The former one is where programmers control how users experience the designs while the latter is where users perceive the design and provides the best patterns for

usability. Furthermore, Loomes' quoted n.d. (as cited in Gentry, 2016) that design is another visual form of communication by knowing how effectively the software conveys information through the designs a programmer has made.

Communication. Possibly the most disregarded aspect of programming has little to do with the mental and manual labour of writing codes. In the workplace programmers need to be able to work with other co-workers and explain to their clients their own works. Programming a clean and effective code is great, but if one is to pair it with strong communication skills by effectively communicating to end users the software one has made, one becomes the rising cream ("10 Skills", 2014). In fact, according to Computer Programming (2020), programmers generally tend to spend most of their time working solo in front of a computer, however, some situations require you to communicate with other programmers and team members, particularly when working on a complex or large projects. Lastly, it is also stated that excellent communication skills will help a programmer to work effectively in the workplace with other to share ideas solutions and ideas efficiently.

Self-reliance. According to 10 Skills (2014), in order to combat certain challenges in programming, there are several things one can do but the first thing is accepting responsibility. Self-reliance is the ability to come up with solutions to problems with as little direct help from outside as possible (Hamm, 2020). It is stated that when one has taken responsibility over something, the information that that one consume will dictate the abilities to exceed its limitations. In fact, according to Moore (2020), self-reliance has a strong theoretical significance in one's own health, happiness and productivity in terms of self-worth, self-expression, self-knowledge, resilience and for self-acceptance that could affect one's own performance in the workplace. Since self-reliance is directly related to

web development through coding skills, it becomes a necessity to also develop this competence to effectively accomplish tasks.

Empathy. As stated by Molenberghs (2017), empathy is the competence to share and understand the emotions of others, as well as their perspective onto something. Having the competence on empathy in the workplace while communicating with other people is an important asset to attain (Mistry, 2020). In fact, she also stated that understanding how other people feels in a certain situation allows people to avoid conflicts, miscommunications and helps to critically evaluate situations and decisions more thoroughly. To summarize, Lee (2020) added that having empathy on your roster of coding skills will help you significantly to land better jobs and to have a good and prospering relationships in the workplace.

Research Literature

Coding has become centralized to businesses that the common mantra around technology experts and researchers is that today's children will need to learn to code or "get coded" (Hargrave, 2018). In addition, the discussion and studies about coding and coding competencies remains abundant. In connection, this presents the discussions of research literatures came from published and unpublished research studies that have direct relevance on the present study. Also, it discusses the objective/s of the study, applicable findings, conclusions and recommendations.

According to Tuomi (2017), digitalization is one of the most promising ways to secure sustainable employment in businesses and is needed to reform the economy by

creating new innovation-related jobs. The application of digital services requires certain competencies. The results of the study indicates certain competencies needed in the application of digital services. They are design skills, logical thinking, problem solving, knowledge on how computers and networks operate, and programming skills. These abilities can be considered as coding skills which are significant in this digitalization. Based on another study by Ibezim & Chukwujekwu (2017), which identifies the computer programming skills required for sustainable employment in Enugu metropolis, Nigeria, it states that 25 hard skills, 19 soft skills and 18 business skills are required for sustainable employment in computer programming jobs. These competencies identified include: ability to program, test and debug programs efficiently and effectively; aptitude to discover and analyze application design considerations for various technologies, aptitude to evaluate users' needs and specifications then develop and implement that software to meet those needs, aptitude to endorse software improvements for clients' existing programs and systems, competence in data mining, open to feedbacks but still confident in personal ideas, adaptive to changes but still remains concentrated on project with upmost priority and has a good sense of judgment. It was concluded that the identified aptitudes should be merged in the curriculum and be emphasized of its importance for sustainability in computer programming jobs as well as in the overall business industry and economy. In another study by Han (2010), which presents the most updated skill requirements for programmer/analyst/web developer, indicates that the programmer/analysts/web developers in the Fortune 500 are projected to justify several roles from computer program writers to technical experts as well as businessmen. In conclusion, skills associated to software, social skills, development and business were exceedingly necessary for

programmer/analysts/web developers by huge corporations. On the other hand, there seems less focus given to aptitudes under the categories of network/architecture, management, hardware and problem solving.

According to Kanbul's study (2017), which aims to reveal the importance of coding education and competencies, it states that it is really vital for students to receive education on coding and enhance their coding competencies in order to be able to acquire 21st-century skills. These skills refers to the wide set of knowledge, skills, character traits and work habits that are believed by— school reforms, educators and employers— to be critically imperative to one's success in today's world; particularly in academic programs, contemporary careers and overall (Edglossary, n.d.). However, there is no attempt to integrate coding education into primary, secondary and university education programs and there is a little number of academic research related with coding education and its competencies in North Cyprus. On the other hand, in the study of Wu et al. (2019), it states that many education systems in Finland, Mainland China, Singapore, Taiwan, and South Korea have recognized the importance of computational thinking, coding skills and are implementing curricular changes to introduce coding education into formal school programs. An essential and critical factor for success involves the training of and support for teachers to teach coding competencies. Thus, their study focuses on understanding the perceptions of teachers towards coding, together with identifying the types of support they received, and their readiness and to teach. The findings of their study specify that educators in Finland, Singapore, Taiwan, and South Korea believe that coding competencies are useful even if students will not work in ICT jobs while Mainland Chinese educators are undecided in this matter. Moreover, according to Hwang et al.'s study (2010) web

development has become a prevalent and dynamic issue in recent years. The swift development of various complex websites not only demonstrates the importance of web development, but also exposes the difficulty of training relevant coding skills which also requires a certain level of cognitive development. The difficulty is due to the lack of facilities such as debugging, online coding and peer assistance to support the learners in improving their cognitive development in web development. The researchers made use of a program which is able to support programming activities with varying difficulty levels of cognition. The results display that the innovative approach is helpful to the students in improving their coding competencies and cognitive development in web development. Furthermore, most of the participants perceived the ease of use and usefulness of the proposed program. Therefore, their study suggests that teachers could integrate web development activities to improve students' coding skills and cognitive development in web development.

According to the study of Burning Glass Technologies (2016), which indicates the importance of learning to code and the value these skills have in the job market, it states that job market demands for these coding competencies have been climbing upward and its breadth and magnitude of employer demands have increased which opened a lot of opportunities for programmers. However, in the study of Xia et al. (2019), states that coding proficiency is essential to software practitioners and web developers. Unfortunately, the understanding on coding skills often translates to vague stereotypes, e.g., "able to write good code". The deprivation of descriptions deter employers from measuring a programmer's coding competencies, and programmers/web developers from improving their coding competencies. Thus, their study focused on identifying these essential coding

skills. The results of their study highlighted a total of 21 important competencies for a programmer/developer to have. In addition, the study of Lee et al. (2013), which intends to identify the knowledge and competencies for web developers by gathering and evaluating job advertisements, concludes that the specific requirements for web developers are proficiency on programming languages such as HTML, CSS, JavaScript, Java, databases, and Internet/Web knowledge/skills for general development and lastly is knowledge on information technology. Additionally, in the study of Dishman (2016), it states that apart from companies in the technology sector, there are growing number of businesses relying on computer programming. No wonder that eight of the top 25 jobs that year are tech positions, according to Glassdoor. The Bureau of Labor Statistics (BLS) projects that employment specifically for computer programmers will increase by 8% to 302,200 positions over the next decade. In fact, in the new survey of Glassdoor indicates that there are seven million job openings in 2015 that required coding skills and programming jobs overall are growing 12% faster than the market average. This new finding is the result of researchers broadening the scope of their analysis to include coding skills across five major job categories: Data Analysis, Information Technology (IT) worker, Web Developers, Engineers and Scientist.

Synthesis

From the preceding studies that the student-researchers have reviewed, it appears that there are certain sets of competencies required by a web developer/programmer to have not just for oneself but also for a sustainable employment. The gathered and analyzed data of essential competencies needed are in accordance to the study's competency parameters,

which are proficiency on code analysis, programming language, design skills, interpersonal skills, self-reliance and empathy. Furthermore, these competencies are proven to be significant in this digitalization and are essentially and progressively demanded by various big and developing corporations. Moreover, it is predicted that in the coming years, coding skills would be certainly an in-demand competency in the workplace. This shows that preparation and training for aspiring students are in need of provision with this matter since these demands should be supplied.

In addition, in order to respond competently to industry prerequisites, it is noteworthy that web education should be given an importance and should provide accommodations in order to satisfy these job demands. In addition, it is really imperative for students to recognize the importance of coding competencies and should receive coding education and enhance their coding skills. In fact, many educational systems have already implemented curricular changes to introduce coding education into formal programs since these countries have recognized the importance of coding skills not just for a specific field but also for the growth of the economy as well. Furthermore, few institutions have now attempted to integrate programs that supports programming activities with varying difficulty levels of cognition which enables students to practice their coding skills. The outcomes from these institutions display that this innovative approach is helpful for the students in improving their coding competencies and cognitive development for web development. Overall, it can now be seen that such competency in the workplace doesn't just apply for coders/programmers but it also transmits unto designers, engineers, architects, basically all jobs that involves technological equipment. To conclude, the importance of

coding competencies should be disseminated, not necessarily just for aspiring web developers, but also for aspiring innovators of this modern world of digitalization.

CHAPTER 3

METHODOLOGY

In this chapter, it presents the used research design of the researchers, together with the instrument used in data gathering, the subjects of the study, the procedure used in gathering the data and the statistical tools to treat and interpret the data.

Research Design

The study used a quantitative research approach for it emphasizes objective statistical, numerical or mathematical measurements to analyze the collected data through the use of polls, questionnaires and surveys, or by controlling pre-dating statistical data using calculating methods (Babbie, 2010). Also, the study made use of descriptive survey research methodology in order to observe, describe and document aspects of a situation as it naturally occurs. With accordance to Hale (2011), descriptive methods have three main types: the observational method, the case-study method and survey method. With conducting the research using survey methods, according to Hale (2011), the people who are part of a research study are entirely guided throughout answering the questions over with interviews or questionnaires. By using this method, the researchers can describe the respondents' answers systematically and statistically evaluate the facts and attributes of a given population.

Respondents of the Study

The respondents of the study are students from both Grade 11 and 12 Technical Vocational and Livelihood Information and Communication Technology of Fidelis Senior High School where the study has taken place at First Asia Institute of Technology and Humanities (FAITH) during the school year 2020-2021. Both Grade 11 and 12 of Technical Vocational and Livelihood Information and Communication Technology students were chosen as the respondents for this study because of their related knowledge and skills and its relation to the study's problem regarding to coding skills for web development. Also, the academic track of the respondents is related to web development industry that will help the researchers to collect accurate data needed for the study. The respondents of the study are fifty (50) in total, exception for the student-researchers. This is from the combination of Grade 11 TVL ICT, who has a total population of thirty-four (34), and Grade 12 TVL ICT, who has a total population of sixteen (16). The chosen procedure for the entire population of the respondents is particularly considered as a total population sampling method where the student-researchers choose to scrutinize the total population on which bears specific set of attributes. (Laerd Dissertation, 2012).

Data Gathering Instrument

As the student-researchers approved and have settle with the validation, they used an adapted survey questionnaire with modification to obtain data for the study. The origin of the statements were from the study of Aaeshim et al. (2009); Wagnild & Young (1993); Abutal et al. (2018); Tang et al. (2016); Bringula et al. (2016); and Yu & Yang (2010). The

student-researchers modified the research questions in order to be applicable to discuss the main point of the study with validation and approbation from teachers to use in survey. According to Snap Surveys (2018), adapted survey questionnaire's layout and content is now viewed with automatic changes to extend survey response valuation. A 5 – point Likert scale was used to interpret the data from survey questionnaire.

Data Gathering Procedure

Before sending out the prepared survey questionnaires to collect data from the chosen respondents, the student-researchers prepared a letter of approval to settle the preliminary needs, settle sudden changes before the gathering process and permit the data gathering processes. The student-researchers asked permissions and consulted to the validators of questionnaires that produce prepared survey questionnaires for the respondents. The student-researchers will ask permission to BES School Principal and Research Advisor of Practical Research II that allows the student-researchers to conduct a survey. Upon approval of the letter, the student-researchers will start distributing the survey questionnaire. When the survey questionnaire has been answered by the respondents, the gathered data will now undergone through tallies and tabulation for the interpretation and analysis of data. By using Statistical Package for the Social Sciences (SPSS) tallies, tabulation and results will be more convenient, less time consuming and accurate results will arise which will help the student-researchers to discuss and present the results with greater understanding, interpretation and further explanation of the findings. After the results and discussion, the student-researchers will rank it to obtain the result that will stood out the most. After presenting and describing the results with careful management, the

student-researchers will summarize the findings. Lastly, with all those careful interpretation and analyzing of the findings, conclusion and recommendations will be established.

Statistical Treatment

The following statistical treatment will be used in order to have solutions to the problems:

Weighted Mean. It was used to know the respondents' perception on the importance of coding skills for web development.

$$\bar{x} = \frac{\sum_{i=1}^n (x_i * w_i)}{\sum_{i=1}^n w_i}$$

Where:

- W** = weighted average
- n** = number of terms to be averaged
- w_i** = weights applied to x values
- x_i** = data values to be averaged

In the computation of the weighted mean, the following weights were utilized:

Table 1

<i>5-point Likert Scale Table</i>		
Option	Range	Interpretation
5	4.21 – 5.00	Very Important
4	3.41 – 4.20	Important
3	2.61 – 3.40	Fairly Important
2	1.81 – 2.60	Slightly Important
1	1.00 – 1.80	Not Important

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APPENDIX A

Letter to the Validator



December 2, 2020

Mark S. Sangcap, MIT
Fidelis IT Department

Good day!

We are the students from Fidelis Senior High School, Grade 12 taking the Technical-Vocational-Livelihood track. As part of our academic endeavour, we will be conducting a quantitative research study entitled: **“The Perception of Fidelis Senior High Technical-Vocational and Livelihood Information and Communication Technology Students on the Importance of Coding Skills for Web Development”**, which aims to gather different perceptions of students regarding the importance of coding skills for web development.

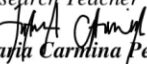
We will gather information for the accomplishment of our study and in line with this, we are humbly requesting for your expertise and assistance in validating our questionnaire. We would be grateful to receive opinions that would support our study with respect to your expertise regarding the validation of our questionnaire.

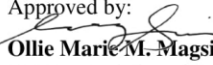
We sincerely give importance and admiration to your participation for the success of our study. Thank you very much and if you have further inquiry, please feel free to contact us through s2019100116@firstasia.edu.ph.

Sincerely,

Abarra, Christian Mark U.
Abutal, Clark Wayne A.
Bagtas, Cloud Neo D.
Gutierrez, Allan R.
Veneracion, Jerwin M.
Villegas, Jan Raynier M.

Noted by

Maria Carmelina Perez
Research Teacher

Maria Carmelina Perez
Inquiries, Investigations, and Immersion Teacher

Approved by:

Ollie Marie M. Magsino
SAC in English and Research, SHS

APPENDIX B

Letter to the Principal



December 2, 2020

Mr. Arnold I. Catapang
BES, Principal

Good day!

We are the students from FAITH Fidelis Senior High School, Grade 12 taking the Technical-Vocational-Livelihood track. As part of our academic endeavour, we would like to request for your permission to allow us to conduct a survey to Grades 11 and 12 TVL-ICT FSH students concerning their perception on the importance of coding skills. The information will be used for the study that we will be conducted entitled: **“The Perception of Fidelis Senior High Technical Vocational and Livelihood Information and Communication Technology Students on the Importance of Coding Skills for Web Development”**. Their responses and identity will be treated with confidentiality for the sake of data privacy and would only be used for our research. Furthermore, a proper disposal of data gathered will be done after the research period.

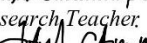
We hope that you will grant our request. Thank you very much and we are looking forward for your favourable response.

Sincerely,

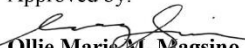
Abarra, Christian Mark U.
Abutal, Clark Wayne A.
Bagtas, Cloud Neo D.
Gutierrez, Allan R.
Veneracion, Jerwin M.
Villegas, Jan Raynier M.

Noted by:


Maria Carmina Perez
Research Teacher


Maria Carmina Perez
Inquiries, Investigations, and Immersion Teacher

Approved by:


Ollie Marie M. Magsino
SAC in English and Research, SHS

APPENDIX C**Research Survey Questionnaire and Letter of Consent****Dear Respondents,**

Good day! We are the students from FAITH Fidelis Senior High School, Grade 12 TVL-ICT taking the Technical-Vocational and Livelihood Information and Communication Technology track. As part of our academic endeavor, we will be conducting quantitative research study titled: **“The Perception of Technical-Vocational and Livelihood Information and Communication Technology on the Importance of Coding Skills for Web Development”**. We are humbly asking for your time and honesty to answer this survey. We assure you that all the information gathered will be confidential and private and will only be used for the study. Furthermore, a proper disposal of data gathered will be done after the research period.

Thank you very much and we are looking forward for your favorable response. If you have further inquiry, please feel free to contact us through (s2019100116@firstasia.edu.ph)

Sincerely,

Abarra, Christian Mark U.
Abutal, Clark Wayne A.
Bagtas, Cloud Neo D.
Gutierrez, Alan R.
Veneracion, Jewin M.
Villegas, Jan Raynier M.

Research Participation Information Form

This is to certify that I am aware of and understand the following conditions for my participation in the study titled: **“The Perception of Technical-Vocational and Livelihood Information and Communication Technology on the Importance of Coding Skills for Web Development”** to be conducted by the aforementioned researchers for their completion of their subject in Practical Research 2 and Inquiries, Investigations, and Immersions.

Signature over Printed Name of Respondent (Optional)

All data will be treated with confidentiality as adherence to the Data Privacy Act of 2012

**Questionnaire**

Name (Optional): _____

Grade/Strand: _____

As a TVL-ICT student, how often do you practice your coding skills?

☐

Everyday

☐

Twice a week

☐

Once a week

☐

Twice a month

☐

Once a month

Directions: Read and answer each statement carefully by putting a check mark (✓) on every statement following the legend below.

5-Very Important

4-Important

3-Fairly Important

2-Slightly Important

1-Not Important

	5	4	3	2	1
In web development, how do you perceive the importance of...					
1. critical thinking that involves analysis, evaluation and reasoning?					
2. proficiently solving and fixing code issues?					
3. being adept at analyzing source codes?					
4. adeptness at problem solving?					
5. debugging and testing codes?					
6. understanding basic tools used in web building (HTML, Javascript, PHP)?					
7. being knowledgeable at Database Management (XAMPP, Mysql, sql, DML)?					
8. understanding different Operating Systems?					
9. learning different programming languages consistently?					
10. understanding framework, algorithms and logic?					
11. creative thinking that involves generating and combining of new ideas?					
12. being proficient in using presentation graphic tools? (Powerpoint, Google Slides)?					
13. using multimedia and creative softwares (Photoshop, Illustration, Animate) proficiently?					
14. understanding color theory?					
15. effectively using CSS for presenting and organizing HTML documents?					
16. speaking with an appropriate volume level - not too loud and not too soft?					
17. talking at a speed which enables anyone to understand what you're trying to relay?					
18. using comprehensible words and expressions that are familiar?					
19. distinguishing the thoughts and viewpoints of others with open-ended questions?					
20. asking other developers to tell you how he/she feels about the point that you're trying to say?					
21. trying to imagine how you would feel in other's place first before criticizing others?					
22. comprehending things easily from the other's point of view?					

All data will be treated with confidentiality as adherence to the Data Privacy Act of 2012



23. trying to look at both side of a disagreement before making a conclusion or decision?					
24. understanding the things that easily make others happy?					
25. understanding easily the cause when someone tells you about an event that made him/her happy?					
26. being able to depend on yourself more than anyone else?					
27. being able to feel that you can handle many things at a time?					
28. being determined?					
29. believing in yourself to get through hard times?					
30. finding your way out in a difficult situation?					

All data will be treated with confidentiality as adherence to the Data Privacy Act of 2012